

$$h = h_P + h_{\perp}$$

$$b = b_P + b_{\perp}$$

Retained h_P

Omitted $h_{\perp}$

Retained b_P

kept interaction

$$\int b_P h_P^{\top}$$

mixed term = 0

$$\int b_P h_{\perp}^{\top} = 0$$

Omitted $b_{\perp}$

mixed term = 0

$$\int b_{\perp} h_P^{\top} = 0$$

entire pairing error

$$\int b_{\perp} h_{\perp}^{\top}$$

Orthogonality

kills both mixed terms.
Only omitted $\times$ omitted remains.

$$\left\| \int b h^{\top} - \int b_P h_P^{\top} \right\|_F \leq \|b - b_P\|_{L^2} \|h - h_P\|_{L^2}$$

History reconstruction error

feedback stability

Trajectory error